

Long-term research directions in theory of computer networks

Based on pre-meeting discussion with Rachit Agarwal, Brighten Godfrey, Robert Kleinberg, and Minlan Yu

Presented by Brighten at the NSF Community Workshop on Long-term Research Directions in Wired Networking

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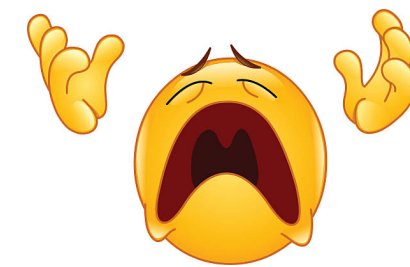
This topic is a little different ...

Theory and algorithms are a tool

- Applied to countless networking problems
- But, not generally *motivating* a new area

So, how is theory relevant in a discussion of future networking research?

- (1) The area is established but the solution bottleneck is a theoretical question
- (2) Intellectual coherence of the domain is crying out for theory



Target audience

- Ourselves stepping back and realizing what we need to do as a community
- Outreach/advertisement to theorists

And please, help us expand the scope covered here!

Four big challenges

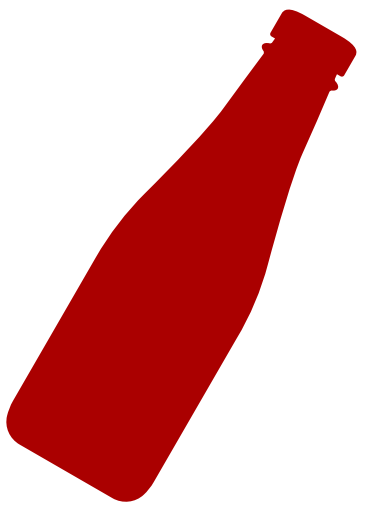
Traffic engineering & topology design

A theory of multipath

Trustworthy ML for systems

A modern theory of congestion control

Traffic engineering and topology



Multi-commodity flow (MCF): It's linear & solvable with an LP optimizer! Done??

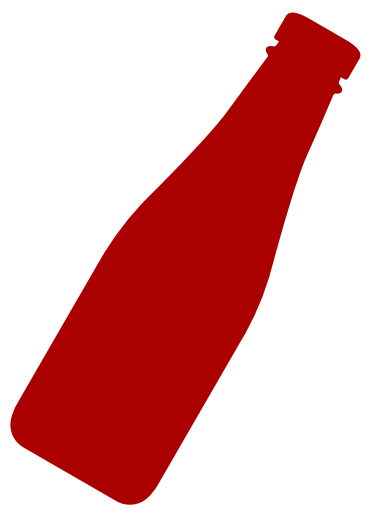
- Nope

1. Faster MCF

- Major clouds using MCF optimizers for 10+ years [B4, SWAN]
- From TEAL paper, SIGCOMM'23: 1.5k nodes, 8k edges, 5.5 hours to optimize in Gurobi
- Limited to approximations for speed (a few minutes) due to huge scale (nodes, policy classes, ...)
- Recent work
 - Pop [SOSP 2021] Solving Large-Scale Granular Resource Allocation Problems Efficiently with POP. https://cs.stanford.edu/~matei/papers/2021/sosp_pop.pdf
 - NCflow [NSDI 2021]: Contracting Wide-area Network Topologies to Solve Flow Problems Quickly. <https://www.usenix.org/conference/nsdi21/presentation/abuzaid>
 - Google PLDP (parallel solver) - could it help?

2. TE for emerging DCNs with directly-networked accelerators

Traffic engineering and topology



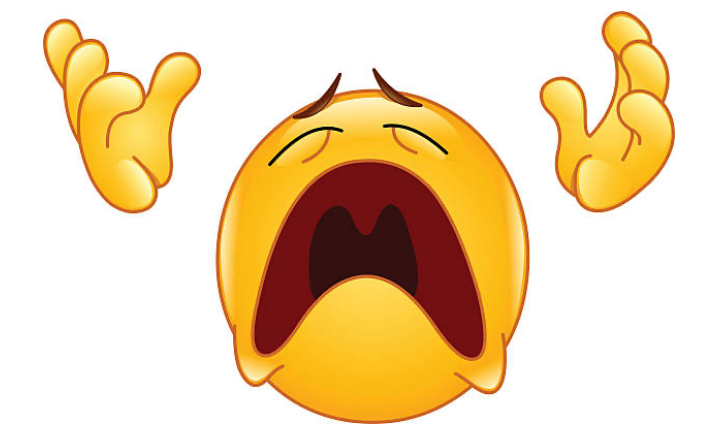
3. Joint MCF + topology design for reconfigurable networks

- Recent theoretical work
 - obliviousness (workload-independence) [Amir et al, STOC 2022 <https://dl.acm.org/doi/10.1145/3519935.3520020>]
 - lower bounds showing it's hard to simultaneously achieve low latency and high network utilization [STOC'21]
- Hierarchy of progressively more complex routing schemes. Oblivious is simplest. Centrally optimized MCF is the most complex. What are the capabilities, limitations of intermediate levels of complexity in routing?
- Exploit predictable ML workloads to plan TE in advance?

4. Joint MCF + topology design + workload placement + congestion control + ...

- Yikes!

A theory of multipath



What happens when endpoints do some of the traffic engineering?

- MPTCP (DC and Internet), MPLS, TeXCP, SD-WAN
- Many emerging multipath transports (for wired + 5G + WiFi, 5G slices, video delivery, ...)
- *We're doing all this and we don't fully understand it* — How many paths are “enough”? What happens when combined with TE? ...with multiple end-hosts simultaneously?

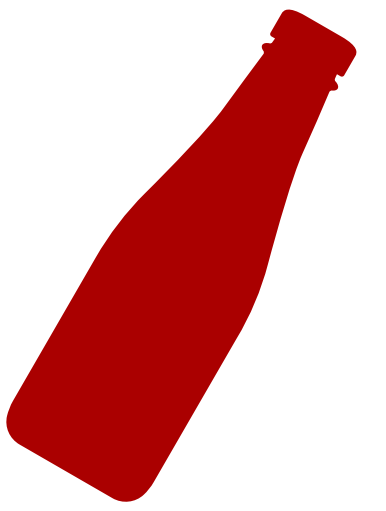
Semi-oblivious routing to the rescue?

- Seems like an apt model of many multipath systems
- Original theory results [Hajiaghayi, Kleinberg, Leighton] pessimistic ... yet works well in practice. Perhaps some key aspect of reality may not be modeled?
- See: <http://www.cs.cornell.edu/~rdk/papers/semiobliv.pdf>, <https://www.cs.cornell.edu/~jnfoster/papers/smores.pdf>, <https://arxiv.org/pdf/2301.06647v2.pdf> ← recent progress! But more to do

Optimizing emerging multipath transport

- Performance & availability critical for real time applications
- Decisions difficult (multiple paths with different properties, unknown future workload)

Trustworthy ML for systems



So you want a “self-driving” cloud...

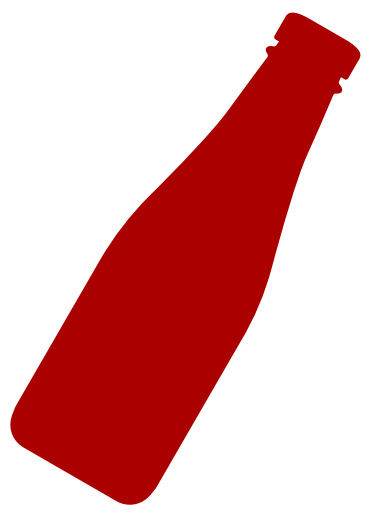
Trustworthy ML for systems

So you want a “self-driving” cloud...

- Self-optimizing, self-monitoring, self-repairing, self-securing, ...



Trustworthy ML for systems



So you want a “self-driving” cloud...

- Cool new idea for a ML-based resource allocator, load balancer, ABR video delivery algorithm, ...
- Works great in your tests, but 1000s of machines/devices/apps are at risk
- **How do we understand robustness of critical infrastructure?**

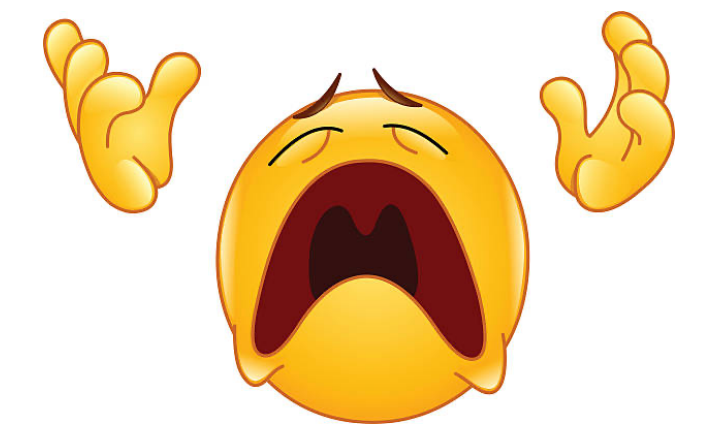
Why is this about ML?

- Not only ML: automated optimization could include heuristics, LP optimizers, etc.
- But especially ML: (1) Uncertainty of behavior outside training regime, (2) Lack of intuitive understanding or theoretical analyzability

Incorporating ML as advice, but with guarantees?

- Perhaps: prediction within ϵ of correct implies chosen decision is within $\epsilon + \delta$ of optimal
- Or perhaps: if advice is good, an NP-hard problem solver will run in poly time
- Could apply this to use predicted properties of the network (e.g., queue lengths) to inform congestion control algorithm decision

Modern theory of congestion control



Traditional theory gave great insight...

- Stability, rate of convergence, utility maximization, fairness, etc...

But it does not describe modern CC algorithms

- We are in the wild west of CC...
- Big gap between EE-style control theory view or Frank Kelly style models vs. modern data-based protocols (dealing with unpredictability, multiplicative increase steps, different utility functions...)

What is the theory of modern congestion control?

- Internet context: BBR, Copa, PCC (regret minimization style), WebRTC for video
- DC context: Swift, HPCC, NDP, Homa, dcPIM
- Metrics? Usefulness of “regret minimization” as a metric? Model diversity of protocols?
- Deployment perf often doesn’t match reality (as evidenced by Pantheon) – what are the models missing?

Discussion during workshop

Need to define the questions so that theorists can make progress on them

Not just optimization: How would an Internet ecosystem evolve?